The trading book capital requirement in practice

Márton Radnai, PhD

Corvinus University of Budapest

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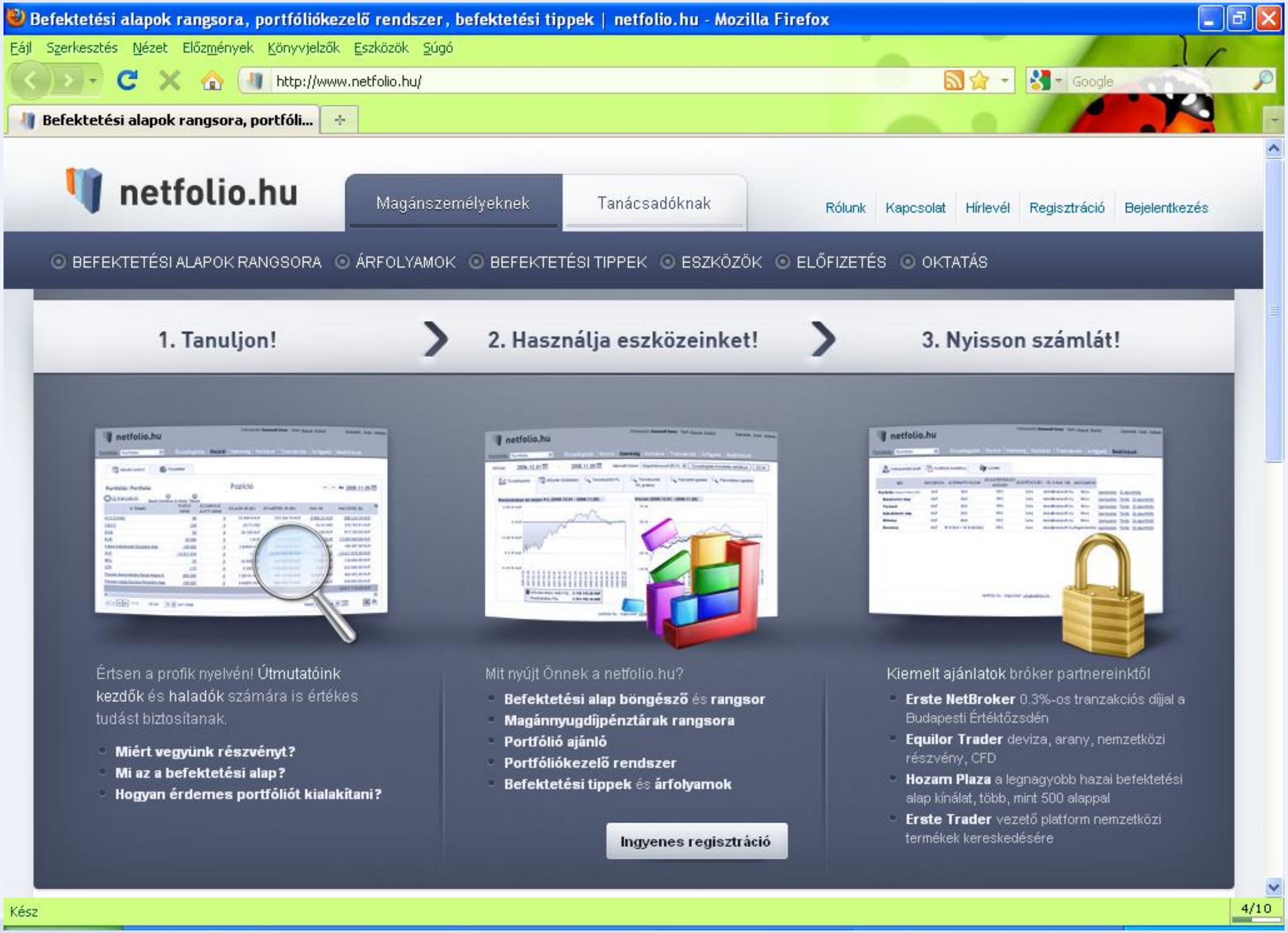
Agenda

- Introduction
- Trading book capital requirement
- Position risk
- Counterparty risk
- Overview of recent changes in trading book regulation (SA-CCR, FRTB)
- Expected changes in future regulation

Introduction

- Ramasoft group – established in 1997, owned by Hungarian individuals
  - 7. fastest growing IT firm in CEE in 2001 (Deloitte & Touche)
  - Main products: Varitron (13 references), Creditron (10 ref.), Pozitron (3 ref.), Oprisk Manager (3 ref.)
  - Distributed products: Atome (11 references), Regnology ESEF (12 references)
  - Main profile: Development and implementation of risk management and reporting systems
  - In 2005 Ramasoft Data Services and Technoligies Ltd. Was established to carry out new developments of the group
  - www.netfolio.hu portal was launched in November 2008
  - Services: Data service, consulting

Netfolio.hu portal



Trading book capital requirements 1.

- History of regulation
  - Basel I. - 1988
  - Basel I. amendment to include market risks – 1996
  - Hungarian regulation – Government decree 244/2000.
    - Slight amendments: Gov. decree 230/2003. (modification)
  - Amendments due to Basel II: 345/2007. (modification) Gov. decree 381/2007. (new, counterparty risk)
    - Slight amendments („Basel 2.5”): Gov. decree 348/2011. (modification)
  - Amendments due to Basel III: Directive 2013/36/EU (CRD IV), EU regulation 2013/575 (CRR), Implementing regulation 2014/680 ITS (Implementing Technical Standards) - direct application
  - Amendments due to „Basel 3.5”: Directive 2019/878 (CRD V) EU regulation 2019/876 (CRR2), Implementing regulation 2021/451 ITS (Implementing Technical Standards) – direct application

Trading book capital requirements 2.

- What is trading book?
  - Positions held with trading intent
    - Positions held for short term (30 days) profits (e.g. long EUR/HUF position)
    - Positions held to serve clients („market making” positions, e.g. government bond inventory, derivative deals concluded with the clients)

Trading book capital requirements 3.

- Elements of capital requirements
  - Position risk
    - General (equity, fixed income, FX, commodities)
    - Individual (only in case of equity and fixed income)
  - Option risks
    - Vega and gamma risk
  - Counterparty risks
    - Settlement risks (free and DVP (delivery versus payment) settlements)
    - Securities financing (repo, securities lending) risks – CRM methods
    - Counterparty risks of OTC derivative deals (forwards, swaps) – CCR methods
  - CVA capital requirement
  - Additional capital requirement of large (single name) risks

Trading book capital requirements 4.

- Applicable methods
  - General position risks
    - Internal model (can be chosen only for one risk factor)
      - VAR based (10 days, 99% confidence interval)
      - 3x „punishment factor”
      - Stressed VAR has to be added (?!)
      - Not used in Hungarian practice as it results in higher capital requirements
    - Standardized method
  - Other risks
    - Standardized method

Position risks 1.

- Position risks
  - Main principle: derivative positions have to be mapped to underlying positions, netted, and these have to be multiplied. Options do not have interest rate risks.
  - Equities: general risk is 8% of net, individual risk is 8% of gross position
  - Fixed income: general risk based on two tables (duration based, maturity based), individual 0-8%
  - FX: 8%, but only the position exceeding 2% of solvency capital has to be taken account (but the total position)
  - Investment units: 40% (!) without mapping, several methods for mapping based on the underlying portfolio

Position risks 2.

- Example 1
  - Position (03.05.2024):
    - 100 OTP
    - 100 2027/A bond
    - 1000 EUR
    - 1000 EUR forward sale for HUF, price 420, maturity 31.12.2024

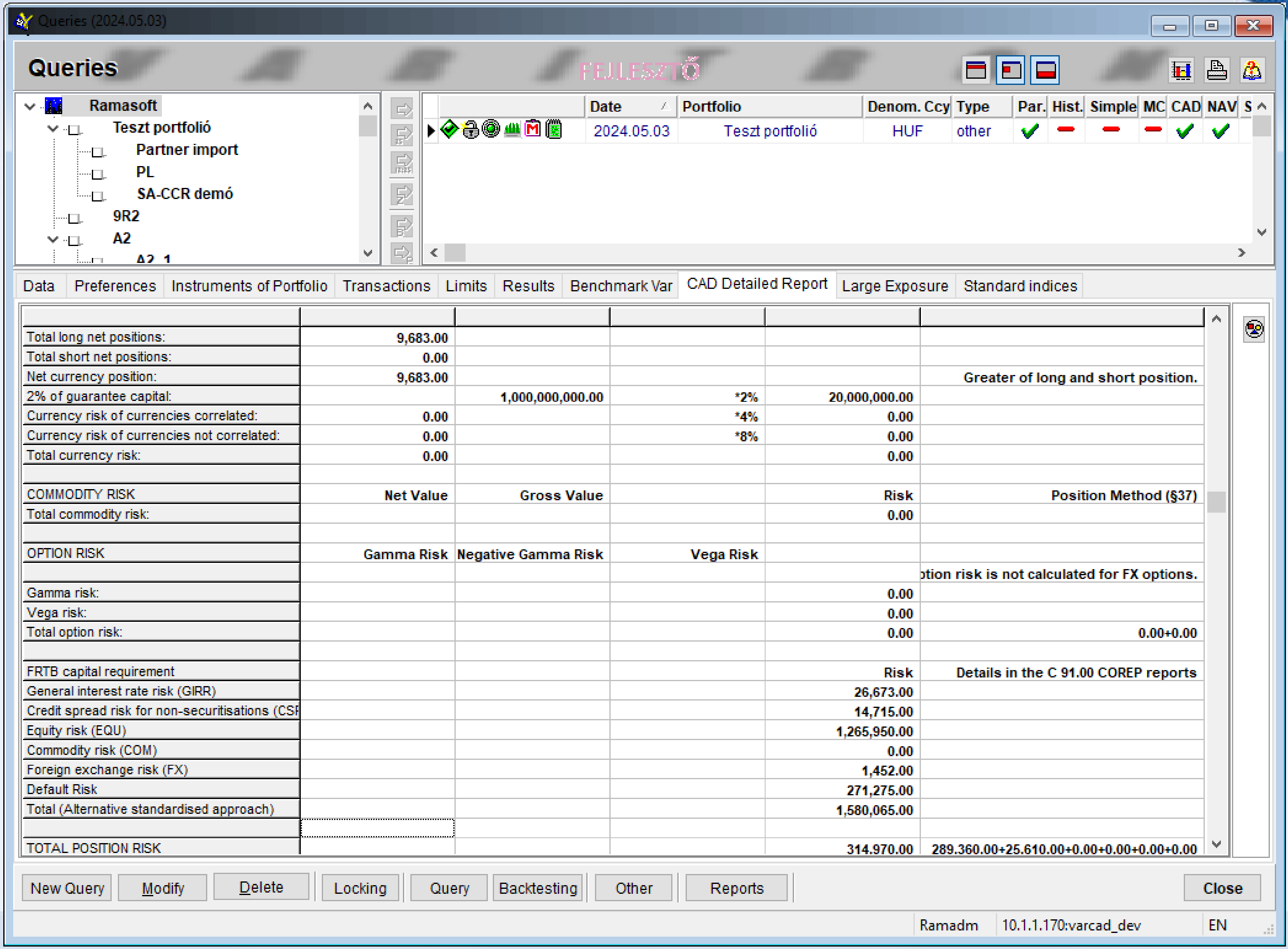
Position risks 3.

- Example (cont.)
  - Mapped position (03.05.2024):
    - 100 OTP $\Rightarrow$ +100 OTP
    - 100 2027/A bond
      - +30 000 HUF 24.10.2024 CF
      - +30 000 HUF 21.10.2025 CF
      - +30 000 HUF 22.10.2026 CF
      - +1 030 000 HUF 25.10.2027 CF
    - 1000 EUR $\Rightarrow$ +1000 EUR/HUF
    - 1000 EUR forward sale for HUF, price 420, maturity 31.12.2024
      - $\Rightarrow$ -1000 EUR/HUF
      - $\Rightarrow$ 420000 HUF 31.12.2024 CF
      - $\Rightarrow$ -1000 EUR 31.12.2024 CF

Position risks 4.

EQUITY RISK	Net position			Weighted position	Method
Single equity risk:	1,808,500.00			144,680.00	1,808,500.00*0.08
General equity risk:	1,808,500.00		1,808,500.00	144,680.00	1,808,500.00*0.08
Total equity risk:				289,360.00	144,680.00+144,680.00
CIUs risk:	0.00			0.00	0.00*0
Equity and CIUs risk:				289,360.00	289,360.00+0.00
INTEREST RATE RISK	Net position			Weighted position	
Single interest rate risk:	894,870.00			0.00	Computed by issuers
General EXPIRATION bas.int.rate risk:	Net position		Weight	Weighted position	
Balanced:	0.00		10%	0.00	
Balanced position-zone 1:	0.00		40%	0.00	
Balanced position-zone 2:	0.00		30%	0.00	
Balanced position-zone 1-2, 2-3:	0.00		30%	0.00	
Balanced position-zones 1-3:	0.00		40%	0.00	
Balanced position-zone 1-3:	0.00		150%	0.00	
Remainder:	25,610.00		100%	25,610.00	
General interest rate risk:				25,610.00	0.00+0.00+0.00+0.00+0.00+0.00+25,610.00
Total interest rate risk:				25,610.00	25,610.00+0.00
FX RISK	Net position	Spot	Future cashflow	Long position	Short position
EUR	9,683.00	389,790.00	-380,107.00	389,790.00	-380,107.00
Total:	9,683.00	389,790.00	-380,107.00	389,790.00	-380,107.00
HUF	3,104,539.00	2,703,749.00	400,790.00	3,104,539.00	0.00
Total + base currency:	3,114,222.00	3,093,539.00	20,683.00	3,494,329.00	-380,107.00
Total long net positions:	9,683.00				

Position risks 5.



Total long net positions:	9,683.00				
Total short net positions:	0.00				
Net currency position:	9,683.00				Greater of long and short position.
2% of guarantee capital:		1,000,000,000.00	*2%	20,000,000.00	
Currency risk of currencies correlated:	0.00		*4%	0.00	
Currency risk of currencies not correlated:	0.00		*8%	0.00	
Total currency risk:	0.00			0.00	
COMMODITY RISK	Net Value	Gross Value		Risk	Position Method (§37)
Total commodity risk:				0.00	
OPTION RISK	Gamma Risk	Negative Gamma Risk	Vega Risk		
					ption risk is not calculated for FX options.
Gamma risk:				0.00	
Vega risk:				0.00	
Total option risk:				0.00	0.00+0.00
FRTB capital requirement				Risk	Details in the C 91.00 COREP reports
General interest rate risk (GIRR)				26,673.00	
Credit spread risk for non-securitisations (CSR				14,715.00	
Equity risk (EQU)				1,265,950.00	
Commodity risk (COM)				0.00	
Foreign exchange risk (FX)				1,452.00	
Default Risk				271,275.00	
Total (Alternative standardised approach)				1,580,065.00	
TOTAL POSITION RISK				314,970.00	289,360.00+25,610.00+0.00+0.00+0.00+0.00

Position risks 6.

- Example 2
  - Position (04.10.2022):
    - One 2028/B government bond
    - How much is the capital requirement in case of maturity and in case of duration based approaches?

Position risks 7.

- Maturity based approach

2028/B goverment bond		Date:		04.10.2022		Maturity based approach			
Date	Time to maturity	Discount factor	Cash Flow	Present Value	Mapped to maturity	Band	Multilplier	Mapped product	Unmapped product
23.03.2023	0,47	0,96	528	504,90		3-6 months	0,40%	2,02	
23.03.2024	1,47	0,87	450	390,80		1-1,9 years	1,25%	4,89	
23.03.2025	2,47	0,79	450	355,01		1,9-2,8 years	1,75%	6,21	
23.03.2026	3,47	0,72	450	322,50		2,8-3,6 years	2,25%	7,26	
23.03.2027	4,47	0,65	450	292,96		4,3-5,7 years	3,25%	9,52	
23.03.2028	5,47	0,59	450	266,06		4,3-5,7 years	3,25%	8,65	
23.03.2028	5,47	0,59	10000	5912,54	8044,78	4,3-5,7 years	3,25%	192,16	261,46
Total				**8044,78**	**8044,78**			**230,70**	**261,46**

- Capital requirement 8044,78x3,25%=261,46

Position risks 8.

- Duration based approach

2028/B goverment bond		Dátum:		04.10.2022		Duration based approach				
Date	Time to maturity	Discount factor	Cash Flow	Present Value	Mapped to duration	Band	Multiplier	Modified duration	Mapped product	Unmapped product
23.03.2023	0,47	0,96	528	504,90		0-1 years	1	0,42	2,14	
23.03.2024	1,47	0,87	450	390,80		1-3,6 years	0,85	1,33	4,43	
23.03.2025	2,47	0,79	450	355,01		1-3,6 years	0,85	2,24	6,77	
23.03.2026	3,47	0,72	450	322,50		1-3,6 years	0,85	3,15	8,64	
23.03.2027	4,47	0,65	450	292,96	8044,78	3,6 years-	0,7	4,06	8,32	241,12
23.03.2028	5,47	0,59	450	266,06		3,6 years-	0,7	4,97	9,26	
23.03.2028	5,47	0,59	10000	5912,54		3,6 years-	0,7	4,97	205,70	
Total				**8044,78**	**8044,78**			**4,28**	**245,26**	**241,12**

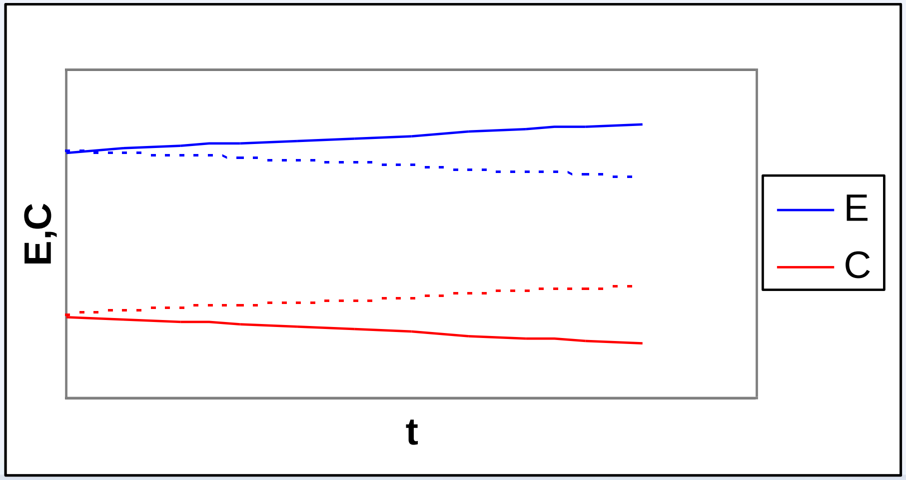
- Capital requirement 8044,78x4,28x0,7%=241,12

Counterparty risks - transactions

- Securities financing transactions – CRM methods
  - Repo, reverse repo
  - Security and commodity lending
  - Margin trading
  - Long settlement transactions
- OTC derivative transactions – CCR methods

Counterparty risks – methods 1.

- Counterparty risk of securities financing transactions – CRM methods
  - Repo, securities lending, margin trading
  - Application of haircuts for the exposure (E) and the collateral (C) as well
    - Standard regulatory haircuts
    - Own calculated haircuts
    - VAR method



Counterparty risks – methods 2.

- Capital requirement = $E^* \cdot Pks \cdot 8\%$

$E^* = E \cdot (1+He) - C \cdot (1-Hc-Hfx) \cdot mm$

E^* adjusted exposure

E (expected) value of exposure

Pks counterparty risk weight

C qualified financial collateral

He, Hc, Hfx haircuts (exposure, collateral, FX)

mm maturity mismatch

- If margining is not done daily then the haircut has to be increased accordingly

Counterparty risk methods

■ CRR	■ CRR2

- Original exposure method
  - nominal x regulatory multiplier
  - can not be chosen if the bank keeps a trading book
- Mark-to-market method
  - Replacement cost + potential future exposure
- Standardized
- $E = 1{,}4 \max\left\{ \begin{matrix} V-C, \\ \sum_j \left(\sum_i RPT_{ij} - \sum_l RPC_{lj}\right) CCRM_j \end{matrix} \right\}$
- For netting sets

- Original exposure method
  - $\mathrm{EAD} = 1.4 \times (\mathrm{RC} + \mathrm{PFE})$
  - RC=max(CMV,0)
  - PFE=nominal x regulatory multiplier
  - Netting agreements are taken into account in RC, but not in PFE, collaterals and margins are not
  - Only small institutions can choose it (TB under 5% and EUR 100M)
- Simplified SA-CCR
  - $\mathrm{EAD} = 1.4 \times (\mathrm{RC} + \mathrm{PFE})$
  - RC=max(CMV,0)
  - Netting agreements are taken into account, collaterals and margins are not
  - Simpler Add-on calculation
  - Medium sized institutions can choose it (under 10% and EUR 300M)
- SA-CCR
  - $\mathrm{EAD} = 1.4 \times (\mathrm{RC} + \mathrm{PFE})$

Counterparty risk methods 2.

- SA-CCR method almost always gives the lowest capital requirement
  - Takes into account value of collaterals and margins when calculating replacements costs
  - Nets potential future exposure
  - Negative replacements cost (that is replaced with a 0 on replacement cost calculation) decreases potential future exposure
- The more clients we have netting agreements with, the higher its advantage is

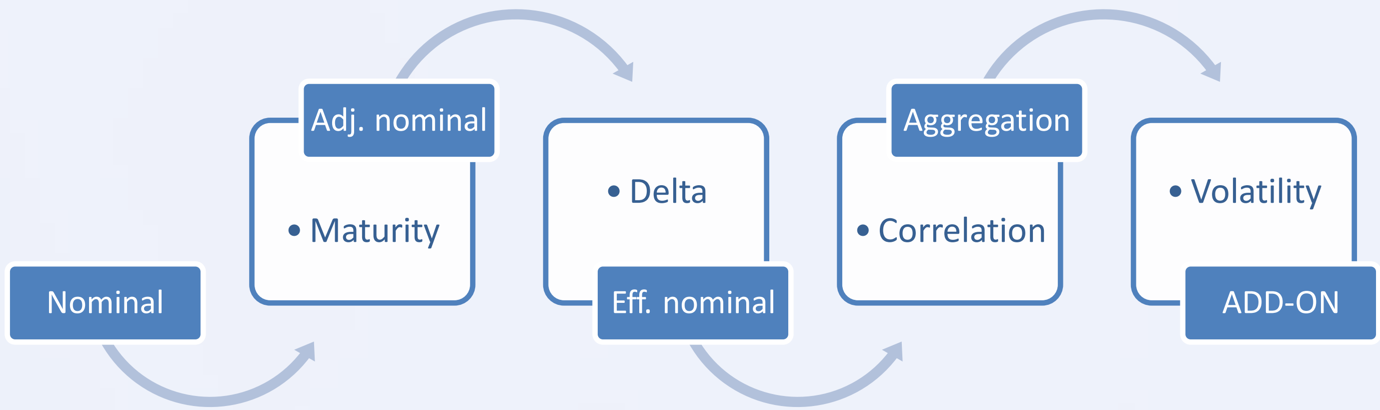
SA-CCR

$$Exposure\ (EAD) = 1.4 \times (RC + PFE)$$

$$Replacement\ cost\ (RC) = \max\{CMV - VM - NICA; TH + MTA - NICA; 0\}$$

$$Potential\ future\ exposure\ (PFE) = m\ \times\ AddOn^{Agg}$$

$$m - if\ RC\ is\ positive\ 1, if\ negative\ between\ 0\ and\ 1$$



SA-CCR 2.

- FX risk

$$AddOn_j^{FX} = \epsilon_j \cdot SF^{FX} \cdot \left|EffNot_j^{FX}\right|$$

ϵ_j- regulatory coefficient of netting set j. (generally 1, if the underlying is volatility or correlation then 5, if spread, then 0.5)

SF^{FX}=4%

$$EffNot_j^{FX} = \sum_{l \in FH_j} Riskposition_l$$

FH – netting set

SA-CCR 3.

- FX risk (cont.)

$$Riskposition = \delta \cdot AdjNot \cdot MF$$

δ – regulatory delta (Black-Scholes, regulatory volatility 15%)

$$\mathrm{AdjNot} = PQ$$

$$MF(no\ margin) = \sqrt{\min(\max\left(M, \frac{10}{252}\right), 1)}$$

M - remaining maturity in years

SA-CCR - example

Portfolio

Data | Preferences | Instruments of Portfolio | Transactions | Limits | Results | Benchmark Var | CAD Detailed Report | Large Exposure | Standard indices

Name	Type	Nominal Value	Ccy	Piece(s)	Value(%)	Start Price	Start	Trading Price	Date of Expir.	Historical cost	NAV
EUR/HUF forward	HFO	-50 000.00	EUR	-50 000.00				352.06	2020.09.25		80 284.85
EUR/HUF forward	HFO	50 000.00	EUR	50 000.00				358.65	2020.09.25		-408 641.65
EUR/HUF forward	HFO	1 000 000.00	EUR	1 000 000.00				331.35	2020.08.31		18 782 321.37
EUR/HUF opció	OE	220 000.00	EUR	220 000.00				347.00	2020.07.01		745 982.30
USD/HUF forward	HFO	-164 683.00	USD	-164 683.00				311.71	2020.05.29		-1 056 422.10
USD/HUF opció	OE	74 385.00	USD	74 385.00				319.15	2020.06.02		178 179.35

Table | Icons | Sum

Data | Preferences | Instruments of Portfolio | Transactions | Limits | Results | Benchmark Var | CAD Detailed Report | Large Exposure | Standard indices

Deal	ID	Partner Name	Type	...	Direction	Nominal Value	Currency	Nominal Value in bas...	Current Value	Currency	Cı
Margin deposit	BP_BANK	BUDAPEST BANK Nyrt.	FUGGETLEN		V				1 295 800	HUF	
Margin agreement	BP_BANK	BUDAPEST BANK Nyrt.				1 295 800	HUF	1 295 800	1 295 800	HUF	
Derivative	BP_BANK	BUDAPEST BANK Nyrt.	EGYEB	D	E	17 603 000	HUF	17 603 000	50 000	EUR	
Derivative	BP_BANK	BUDAPEST BANK Nyrt.	EGYEB	D	V	17 932 500	HUF	17 932 500	50 000	EUR	
Derivative	BP_BANK	BUDAPEST BANK Nyrt.	EGYEB	D	V	331 350 000	HUF	331 350 000	1 000 000	EUR	
Derivative	BP_BANK	BUDAPEST BANK Nyrt.	OPCIOPUT	D	V	76 340 000	HUF	76 340 000	220 000	EUR	
Derivative	BP_BANK	BUDAPEST BANK Nyrt.	EGYEB	D	E	51 333 338	HUF	51 333 338	164 683	USD	
Derivative	BP_BANK	BUDAPEST BANK Nyrt.	OPCIOPUT	D	V	23 739 973	HUF	23 739 973	74 385	USD	

SA-CCR – example 2.

Replacement cost (RC)

Transaction	Netting set	CMV	VM	NICA	TH	MTA	CMV-VM-NICA	TH+MTA-NICA	RC - SA-CCR	RC - Simplified SA-CCR, Original exposure method	RC – Without netting agreement
FWD96634	EUR/HUF	80 285									80 285
FWD95794	EUR/HUF	-408 642									0
FWD96118	EUR/HUF	18 782 321									18 782 321
FXO56105	EUR/HUF	745 982									745 982
FWD94866	USD/HUF	-1 056 422									0
FXO56279	USD/HUF	178 179									178 179
Budapest Bank Nyrt. Total		**18 321 704**	**0**	**1 295 800**	**1 295 800**	**1 295 800**	**17 025 904**	**1 295 800**	**17 025 904**	**18 321 704**	**19 786 768**

Add-on

Transaction	Netting set	Nominal value	Adjusted nominal value	Delta	Mf	Risk position	Total add-on	y	CMV-VM-NICA	z	Multiplier	Add-on
FWD96634	EUR/HUF	17 449 500	17 449 500	-1	0,298807	-5 214 035	-208 561					0
FWD95794	EUR/HUF	17 449 500	17 449 500	1	0,298807	5 214 035	208 561					0
FWD96118	EUR/HUF	348 990 000	348 990 000	1	0,298807	104 280 707	4 171 228					0
FXO56105	EUR/HUF	76 777 800	76 777 800	-0,44208	0,298807	-10 142 164	-405 687					0
FWD94866	USD/HUF	52 380 722	52 380 722	-1	0,298807	-15 651 734	-626 069					0
FXO56279	USD/HUF	23 659 637	23 659 637	-0,54497	0,298807	-3 852 774	-154 111					0
EUR/HUF total		460 666 800	460 666 800			94 138 543	3 765 542					0
USD/HUF total		76 040 359	76 040 359			-19 504 508	780 180					0
Budapest Bank Nyrt. Total		**536 707 159**	**536 707 159**			**74 634 035**	**4 545 722**	**8 636 872**	**17 025 904**	**17 025 904**	**1**	**4 545 722**

SA-CCR – example 3.

SA-CCR risk weighted exposure (RWE) – according to our interpretation

Transaction	**Netting set**	**Distributed RC**	**Distributed add-on**	**Exposure**	**Risk weight**	**RWE**
FWD96634	EUR/HUF	69 082	0	96 715	100%	96 715
FWD95794	EUR/HUF	0	179 312	251 037	100%	251 037
FWD96118	EUR/HUF	16 161 608	3 586 230	27 646 973	100%	27 646 973
FXO56105	EUR/HUF	641 895	0	898 653	20%	179 731
FWD94866	USD/HUF	0	626 069	876 497	20%	175 299
FXO56279	USD/HUF	153 317	154 111	430 399	20%	86 080
Budapest Bank Nyrt. Total		**17 025 904**	**4 545 722**	**30 200 274**		**28 435 835**

SA-CCR – example 4.

C 34.03 FEJLESZTŐ

Date of Query: 2020.05.21 Portfolio Name: Teszt portfolió

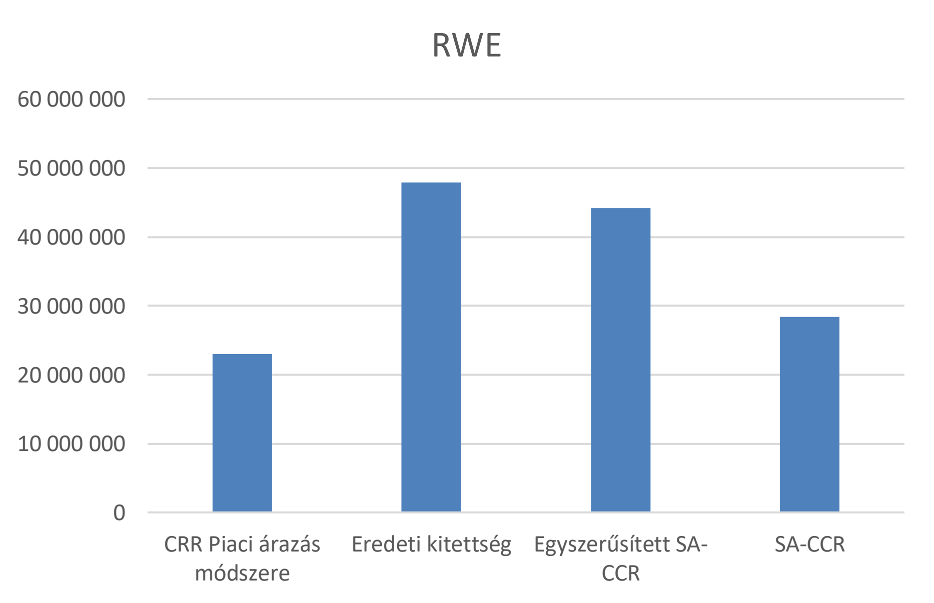
C 34.03(s0001) | C 34.03(s0002)

		Currency	Second currency in pair	Number of transactions	Notional amounts	Current market value (CMV), positive	Current market value (CMV), negative	Add-on
		0010	0020	0030	0040	0050	0060	0070
Összesen	0010			6.0000	536 707 159.0000	19 786 767.0000	-1 465 064.0000	4 545 722.0
Ebből: 2 kockázati kategóriához hozzárendelt	0020			0.0000	0.0000	0.0000	0.0000	
Ebből: 3 kockázati kategóriához hozzárendelt	0030			0.0000	0.0000	0.0000	0.0000	
Ebből: Több mint 3 kockázati kategóriához hozzárendelt	0040			0.0000	0.0000	0.0000	0.0000	
Kamatlábkockázat	0050			0.0000	0.0000	0.0000	0.0000	0.0
Ebből: Kizárólag a kamatlábkockázati kategóriához hozzárendelt	0060			0.0000	0.0000	0.0000	0.0000	
Ebből: Legnagyobb pénznem	0070			0.0000	0.0000	0.0000	0.0000	0.0
Ebből: Második legnagyobb pénznem	0080			0.0000	0.0000	0.0000	0.0000	0.0
Ebből: Harmadik legnagyobb pénznem	0090			0.0000	0.0000	0.0000	0.0000	0.0
Ebből: Negyedik legnagyobb pénznem	0100			0.0000	0.0000	0.0000	0.0000	0.0
Ebből: Ötödik legnagyobb pénznem	0110			0.0000	0.0000	0.0000	0.0000	0.0
Devizaárfolyam-kockázat	0120			6.0000	536 707 159.0000	19 786 767.0000	-1 465 064.0000	4 545 722.0
Ebből: Kizárólag a devizaárfolyam-kockázati kategóriához hozzárendelt	0130			6.0000	536 707 159.0000	19 786 767.0000	-1 465 064.0000	
Ebből: Legnagyobb devizapár	0140	EUR	HUF	4.0000	460 666 800.0000	19 608 588.0000	-408 642.0000	3 765 542.0
Ebből: Második legnagyobb devizapár	0150	USD	HUF	2.0000	76 040 359.0000	178 179.0000	-1 056 422.0000	780 180.0
Ebből: Harmadik legnagyobb devizapár	0160			0.0000	0.0000	0.0000	0.0000	0.0
Ebből: Negyedik legnagyobb devizapár	0170			0.0000	0.0000	0.0000	0.0000	0.0
Ebből: Ötödik legnagyobb devizapár	0180			0.0000	0.0000	0.0000	0.0000	0.0
Hitelkockázat	0190			0.0000	0.0000	0.0000	0.0000	0.0
Ebből: Kizárólag a hitelkockázati kategóriához hozzárendelt	0200			0.0000	0.0000	0.0000	0.0000	
Egy alaptermékes ügyletek	0210			0.0000	0.0000	0.0000	0.0000	
Több alaptermékes ügyletek	0220			0.0000	0.0000	0.0000	0.0000	
Részvénypiaci kockázat	0230			0.0000	0.0000	0.0000	0.0000	0.0
Ebből: Kizárólag a részvénypiaci kockázati kategóriához hozzárendelt	0240			0.0000	0.0000	0.0000	0.0000	
Egy alaptermékes ügyletek	0250			0.0000	0.0000	0.0000	0.0000	
Több alaptermékes ügyletek	0260			0.0000	0.0000	0.0000	0.0000	

SA-CCR – example 5.

Risk weighted exposure (RWE) – comparison of different methods

Method	RC	PFE	Exposure	RWE
CRR Mark-to-market method	19 786 768	5 182 988	24 969 755	23 019 120
Original exposure method	18 321 704	21 468 286	55 705 985	47 901 325
Simplified SA-CCR	18 321 703	13 930 102	45 152 527	44 227 150
SA-CCR	17 025 904	4 545 722	30 200 274	28 435 835



Overview of FRTB 1.

- Fundamental review of trading book (FRTB)
  - Basel – three consultation papers (2012, 2013, 2014), final recommendation was issued in 2016
  - Quantitative Impact Study in 2015, intense lobbying against it afterwards
  - EU – CRR modification proposal of November 2016 included it (but was not accepted)
  - Basel –2 more consultation papers (2017, 2018), amended recommendation issued in 2019
  - Included in CRDV/CRR2 only as a reporting requirement for large banks, in CRDVI/CRR3 proposed as capital requirement

Overview of FRTB 2.

- Aim
  - Decrease regulatory arbitrage
  - Increase risk sensitivity
  - Standardize calculation methods
- Changes
  - Standardized method – different methods for risk factors are deleted, calculation method is same for all risk factors
  - Internal model – based on expected shortfall instead of VAR

Overview of FRTB 3.

■ BIII standardised	■ FRTB alt. standardised

- Fixed income
  - *Maturity, duration based*
- Commodity
  - Staircase, Simple
- FX
  - *8 % of open FX position*
- Equity
  - *Gross, Net 8%*
- Gamma, Vega
  - $\left[\frac{1}{2}\gamma \times (S \times \sigma)^2 \times PV; 0\right]^-$
  - $v \times 100 \times \sigma PV$

- SENSITIVITY BASED METHOD

$$SBM = Delta + Vega + Curvature$$

$$S = \frac{c(P+\delta) - c(P)}{\delta}$$

$$\boldsymbol{K_b = \sqrt{\sum_{k=1}^{n} WS_k^2 + \sum_{k=1}^{n-1}\sum_{k+1}^{n} \rho_{kl} WS_k WS_l}}$$

$$\boldsymbol{Delta\ risk = \sqrt{\sum_{b=1}^{m} K_b^2 + \sum_{b=1}^{m-1}\sum_{c=b+1}^{m} \gamma_{bc} S_b S_c}}$$

b – bucket
S$_k$ – sensitivity
W – weight, $\boldsymbol{\rho_{kl}}$ – correlation between factors k and l
$\boldsymbol{\gamma_{bc}}$– correlation between buckets b and c

- RESIDUAL RISK ADDON
  $RRAO = Notional \times$ Y %
- JUMP TO DEFAULT
  $JTD = [LGD \times notional + P\&L, 0]^+$

FRTB - example

Portfolio

Name	Type	Nominal value	Ccy.	Quantity	Maturity	Time to maturity (years)	Counterparty
HUF cashflow	CF	1 000 000 000,00	HUF	1 000 000 000,00	2021.11.30	0,25	Hungarian state
HUF cashflow	CF	1 000 000 000,00	HUF	1 000 000 000,00	2031.08.31	10,00	Hungarian state
EURO cashflow	CF	1 000 000,00	EUR	1 000 000,00	2026.08.31	5,00	Commerzbank AG

General interest rate risk brackets and weights (to be divided by $\sqrt{2}$)					
Maturity	0.25 years	0.5 years	1 years	2 years	3 years
Risk weight	1.7%	1.7%	1.6%	1.3%	1.2%
Maturity	5 years	10 years	15 years	20 years	30 years
Risk weight	1.1%	1.1%	1.1%	1.1%	1.1%

Ccy.	Maturity bracket	quantity	net asset value	sensitivity	risk weight	weighted sensitivity
HUF	1	1 000 000 000,00	996 015 936,00	248 321 781,00	1,2021%	2 985 030,00
HUF	7	1 000 000 000,00	757 075 815,00	7 574 906 508,00	0,7778%	58 918 945,00
EUR	6	1 000 000,00	354 420 462,00	1 773 073 326,00	0,7778%	13 791 274,00

FRTB – example 2.

Bucket sensitivity

$$K_b = \sqrt{\sum_k WS_k^2 + \sum_k \sum_{l \neq k} \rho_{kl} WS_k WS_l}$$

Correlation

$$\rho_{kl} = max\left[e^{-\theta \frac{|T_k - T_l|}{\min(T_k;T_l)}}; 40\%\right] \qquad \theta = 3\%$$

Delta risk

$$K_S = \sqrt{\sum_b K_b^2 + \sum_b \sum_{c \neq b} \gamma_{bc} S_b S_c}$$

Correlation (HUF) γ_{bc}=50%

FRTB – example 3.

Category sensitivity $K_b = \sqrt{2\ 985\ 030^2 + 58\ 918\ 945^2 + 2 * 0{,}40 * 2\ 985\ 030 * 58\ 918\ 945}$ $= 60\ 175\ 180$

Correlation $\rho_{kl} = 40\%$

Delta risk $\sqrt{60\ 175\ 180^2 + 13\ 791\ 274^2 + 2 * 0{,}50 * 60\ 175\ 180 * 13\ 791\ 274}$ $= 68\ 125\ 942$

Correlation (HUF) γ_{bc}=50%

FRTB – example 4.

Scenarios

$$\rho_{kl}^{weak} = max\,(2 \times \rho_{kl} - 100\,\%; 75\,\% \times \rho_{kl}) = 0{,}75 \cdot 0{,}40 = 0{,}3$$

$$\gamma_{bc}^{weak} = max\,(2 \times \gamma_{bc} - 100\,\%; 75\,\% \times \gamma_{bc}) = 0{,}75 \cdot 0{,}50 = 0{,}375$$

$$\rho_{kl}^{strong} = min(100\%; 125\% \times \rho_{kl}) = 1{,}25 \cdot 0{,}40 = 0{,}5$$

$$\gamma_{bc}^{strong} = min(100\%; 125\% \times \gamma_{bc}) = 1{,}25 \cdot 0{,}50 = 0{,}625$$

Delta risk	Weak	66 455 832
	Strong	70 097 039
Capital requirement	Maximum	70 097 039

FRTB – example 5.

C 91.00

FEJLESZTŐ

C 91.00

Date of Query: 2021.08.31 Portfolio Name: Teszt portfolió

C 91.00

C 91.00 - Alternative Standardised Approach: Summary (MKR ASA SUM)					
		itions subject to sensitivities-based met			
		Unweighted delta sensitivities	nweighted delta sensitiviti	Unweighted delta sensitivities	Own funds requirements under the different scenarios
					Low correlation scenario
		Positive	Negative	Net sensitivities per risk class	Delta Risk
		0010	0020	0030	0040
Total (Alternative standardised approach)	0010	9 950 722 077.0000	0.0000	9 950 722 077.0000	119 618 901.0000
0019 Sensitivity-based method	0015				
General interest rate risk (GIRR)	0020	9 596 301 615.0000	0.0000	9 596 301 615.0000	66 455 832.0000
Credit spread risk for non-securitisations (CSR)	0030	0.0000	0.0000	0.0000	0.0000
Credit spread risk for securitisation not included in the alternative correlation trading portfolio	0040	0.0000	0.0000	0.0000	0.0000
Credit spread risk for securitisation included in the alternative correlation trading portfolio (AC	0050	0.0000	0.0000	0.0000	0.0000
Equity risk (EQU)	0060	0.0000	0.0000	0.0000	0.0000
Commodity risk (COM)	0070	0.0000	0.0000	0.0000	0.0000
Foreign exchange risk (FX)	0080	354 420 462.0000	0.0000	354 420 462.0000	53 163 069.0000
Default risk	0085				
Non-securitisations	0090				
Securitisation not included in the alternative correlation trading portfolio (non-ACTP)	0100				
Securitisation included in the alternative correlation trading portfolio (ACTP)	0110				
Residual risk	0115				
Exotic underlyings	0120				
Other residual risk	0130				

Close

Ramadm ramadev:varcad_dev EN

FRTB – example 5.

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Date of Query: 2021.08.31 Portfolio Name: Teszt portfolió

C 91.00

Own funds requirements under the different scenarios	quirements under the differe	equirements under the differe	unds requirements under the different sce	ns subject to defa	s subject to def	s subject to resic		
High correlation scenario	High correlation scenario	High correlation scenario	High correlation scenario	ıp-to-default (JTD	p-to-default (JTD			
Delta Risk	Vega Risk	Curvature Risk	Total	Long	Short	ross notional valu	Own funds requirements	Total risk exposure amount
0120	0130	0140	0150	0160	0170	0180	0190	0200
123 260 108.0000	0.0000	0.0000	123 260 108.0000				123 260 108.0000	1 540 751 350.0000
70 097 039.0000	0.0000	0.0000	70 097 039.0000				70 097 039.0000	
0.0000	0.0000	0.0000	0.0000				0.0000	
0.0000	0.0000	0.0000	0.0000				0.0000	
0.0000	0.0000	0.0000	0.0000				0.0000	
0.0000	0.0000	0.0000	0.0000				0.0000	
0.0000	0.0000	0.0000	0.0000				0.0000	
53 163 069.0000	0.0000	0.0000	53 163 069.0000				53 163 069.0000	
				0.0000	0.0000		0.0000	
				0.0000	0.0000		0.0000	
				0.0000	0.0000		0.0000	
						0.0000	0.0000	
						0.0000	0.0000	

Close

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Overview of FRTB 4.

- Simplified standardised method – does not appear in CRR2, only in CRR3

$$CR = CR_{IRR} \cdot 1{,}3 + CR_{EQ} \cdot 3{,}5 + CR_{FX} \cdot 1{,}2 + CR_{COMM} \cdot 1{,}9$$

where
CR – capital requirement
$CR_{IRR,EQ,FX,COMM}$ – the Basel II capital requirement of the risk factor (with non delta risk capital requirements of options)

- Capital requirement will increase, especially for banks with large equity risks

Expected changes in regulation

- CRR3/CRDVI – expected to come into effect: 1 January 2025
- Market risk
  - FRTB alternative standardized method not only for reporting but for capital calculation as well
  - Simplified standardized method that calculates capital requirements from current requirements using different multipliers for each risk category
    - Can be chosen by small banks (size of trading book is smaller than EUR 500 M and 10% of balance sheet)
  - Start date can be adjusted…
  - New reports (C24, C92-99)
- Counterparty risk
  - No changes
- CVA
  - New methods (simplified as well)
  - New reports (C25, C92.02)

Suggested readings (in Hungarian)

- Radnai, M. – Vonnák Dzs.: Handbook of capital adequacy (Alinea, 2010)
  - Chapter 2. – Market risk
  - Chapter 3.4. – Counterparty risk
- Bóta, N. - Radnai, M. – Vonnák Dzs.: Handbook of capital adequacy volume II. (Alinea, 2019)
  - Subsection 9.2. – Market risk
  - Subsection 9.3.4. – Counterparty risk
  - Subsection 9.10.1. – Expected changes in market risk
- The two volumes can be purchased at a discount in our office (HUF 5000 in total instead of HUF 6950)

Köszönjük a figyelmet!

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More information

http://www.ramasoft.hu

Márton Radnai, PhD

1074 Budapest, Dohány u. 12-14.

Tel: 473-1219

Fax: 269-3209